

Stat 3113: Lesson 0

MOTIVATION



The Basic Idea

- Analyzing scientific data is **challenging** because data are subject to **random variation** or variability.
- So how can one **draw conclusion** from the results of an experiment when those results could have come out differently?
- The methods of Statistics allow scientists and engineers to **design valid experiments** and to **draw reliable conclusions** from the data they produce.



The Basic Idea

- For better or worse, huge amount of data are collected about our opinions and our lifestyle, for purposes ranging from creation of more effective marketing campaigns to the development of social policies designed to improve our way of life.
- The basic idea behind all statistical methods of data analysis is to make inferences about a population by studying a relatively small sample chosen from it.

Motivating Example

A machine used in clutch systems made 2000 steel balls with a diameter specification of 0.65 ± 0.03 . A sample of 80 balls showed 90% met the specification. However, this sample might not perfectly reflect the entire batch. So, while 90% in the sample meet the spec, it's likely the overall proportion in the 2000 balls could vary from this figure.

- What the engineer conducting this study needs to know is just how large that difference is likely to be.
- For example, is it plausible that the population percentage could be as high as 95% or as low as 80%?

Motivating Example

- More specifically, the engineer might want to address the following questions on the basis of these sample data.
 1. How large is a typical difference for this kind of sample?
 2. Having observed that 90% of the the sample balls were good, can we specify the percentage of acceptable balls in the population as an interval of the form $90\% \pm x\%$, where x is calculated to provide reasonable certainty that the true population percentage is in the interval. So how should x be calculated?
 3. To avoid recalibration, how certain can he be that at least 85% of the 2000 balls produced are good?
- This course is devoted to addressing questions like these.

Motivating Example

- For example,
 - The first question involves computing the standard deviation, a topic covered in detail in Chapter 3.
 - For the second question, we'll explore the construction of confidence intervals, a skill developed in Chapter 5.
 - The third question introduces hypothesis testing, a concept to be studied comprehensively in Chapter 6.
- Subsequent chapters will delve into other crucial statistical topics, offering an engaging and informative journey throughout the semester.

Motivating Example

Furthermore,

- The engineer in our example may want to know how the amount of carbon in the steel balls is related to the compressive strength.
 - ✓ Correlation and Regression → Chapters 2 & 8
- It may also be important to determine how to adjust the manufacturing process with regard to several factors, in order to produce optimal results.
 - ✓ Factorial Experiments → Chapter 9
- Finally, the engineer will need to develop a plan for monitoring the quality of the product manufactured by the process.
 - ✓ Statistical quality control → Chapter 10

Reading

Chapter 1 - Introduction



Next Topic



Introductory Material Part 1



Chapter 1 – Section 1.1